

CVE-2022-23535: LiteDB JSON Object Injection RCE

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Content

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9.8

CVSS Score

3.1

Basic Information

CVE ID

[CVE-2022-23535](#)

GHSA ID

[GHSA-3x49-g6rc-c284](#)

EPSS Score [Info](#)

60.22%

CWE

[CWE-502](#)

Published

2/24/2023

Updated

3/6/2023

KEV Status [Info](#)

No

Technology

[\[image: Technology\]](#)C#

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[CVE-2022-23535](#)

CVE-2022-23535: LiteDB may deserialize bad JSON on object type using _type

Impact

LiteDB use a special field in JSON documents to cast diferent types from `BsonDocument` do POCO classes. When instance of an object are not the same of class, `BsonMapper` use a special field `_type` string info with full class name with assembly to be loaded and fit in your model. If your end-user can send to your app a plain JSON string, deserialization can load an unsafe object to fit in your model.

Patches

Version `>= 5.0.13` add some basic fixes to avoid this, but is not 100% guaranteed when using `Object` type Next major version will contains a allow-list to select what king of Assembly can be loaded

Workarounds

- Avoid users send to your app a JSON string to be direct insert/update into database
- Avoid use classes with `Object` type - try use an interface when possible

If your app send a plain JSON string to be insert/update into database, prefer this:

```
// Bad
public class Customer {
    public int Id { get; set; }
    public string Name { get; set; }
    public Object AnyData { get; set; } // <= Avoid use `Object` base type
}

// Good
public class Customer {
    public int Id { get; set; }
    public string Name { get; set; }
    public IDictionary<string, string> AnyData { get; set; } // Will accept only
key/value strings
}
```

References

See this workaround fix on this commit:

<https://github.com/mbdavid/LiteDB/commit/4382ff4dd0dd8b8b16a4e37dfd29727c5f70f93f>
([GitHub Advisory](#))

Show more

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Is this CVE running in your environment?

Easily map the attack path and prioritize which CVEs are a threat to your organization

[Validate Exposure!\[\]\(05be7c7a8995decd503647c99211f7c2_img.jpg\)](#)

Technical Details

CVSS Vector

CVSS:3.1/AV:N/AC:L/PR:N/UI:N/S:U/C:H/I:H/A:H

| Package Name | Ecosystem | Vulnerable Versions | First Patched Version | | LiteDB |
nuget | < 5.0.13 | 5.0.13 | |

Vulnerability Intelligence

Miggo AI

Root Cause Analysis

The vulnerability stems from how BsonMapper handles '_type' during deserialization. The commit 4382ff4 adds two critical checks: 1) IsAssignableFrom validation to ensure type compatibility, and 2) explicit blocking of System.Diagnostics.Process. The original code (shown in the diff) lacked these checks, making the Deserialize method vulnerable to type confusion attacks. The function's responsibility to resolve and validate types based on untrusted '_type' values directly matches the described exploit scenario where malicious types could be loaded via user-controlled JSON.

Vulnerable functions



Only Mio us*rs **n s t*is s**tion

Unlock WAF rules for this CVE

Generate vendor-ready rules for the observed attack patterns, plus reasoning and safe deployment guidance

[Get WAF rules!\[\]\(c724c83fe216b2427610afdbd31f92cc_img.jpg\)](#)

WAF Protection Rules

WAF Rule

[image: Waf Protection]

[illegible]

Reasoning

[image: Waf Protection]

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